Quantum Information Theory
Lesson 5 Bonus: CHSH Inequality

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Contents

Postulates of Quantum Mechanics

1. The momentary state of an isolated system is a vector $|\psi(t)\rangle \in \mathcal{H}(S)$ in a complex Hilbert space $\mathcal{H}(S)$;
2. Measurable physical quantities are described by Hermitian operators (observables, $\mathbf{O} = \sum_i \lambda_i |\psi_i\rangle \langle\psi_i|$);
3. Measuring an observable produces one of its eigenvalues;
4. Probability of measuring λ is $\sum_{\lambda_i=\lambda} \langle\psi_i \mid \psi(t)\rangle \langle\psi(t) \mid \psi_i\rangle$;
5. When λ is observed, the post-measurement state is

$$|\psi_\lambda(t+1)\rangle \propto \sum_{\lambda_i=\lambda} |\psi_i\rangle \langle\psi_i \mid \psi(t)\rangle ;$$

6. The time evolution of $|\psi(t)\rangle$ is governed by Schrödinger's equation, $\imath\hbar\frac{d}{dt}|\psi(t)\rangle = \mathbf{H}(t)|\psi(t)\rangle$, where $H(t)$ is the observable associated with the total energy (Hamiltonian).

Observables and Density Matrices

We know that if we have a random quantum state

$$|\mathsf{v}\rangle \sim \begin{pmatrix} |\xi_1\rangle & |\xi_2\rangle & \dots & |\xi_n\rangle \\ p_1 & p_2 & \dots & p_n \end{pmatrix}$$

this corresponds to density matrix $\rho = \sum_{k=1}^{n} p_k |\xi_k\rangle \langle\xi_k| \in \mathcal{B}(\mathbb{C}^d)$.
Let's measure upon $|\mathsf{v}\rangle$ an observable $\mathbf{O} = \sum_{i=1}^{d} \lambda_i |\psi_i\rangle \langle\psi_i|$.
There are as many outcomes as distinct eigenvalues of $\mathbf{O}$.
It is convenient to group terms corresponding to the same λ.
Writing $\mathbf{O} = \sum_\lambda \lambda \mathbf{E}_\lambda(\mathbf{O})$, where $\mathbf{E}_\lambda(\mathbf{O}) = \sum_{\lambda_i=\lambda} |\psi_i\rangle \langle\psi_i|$ is the projector on the λ-eigenspace of $\mathbf{O}$, the probability to observe λ is

$$\begin{aligned}
\Pr[\mathsf{x} = \lambda] &= \sum_{k=1}^{n} \Pr[\mathsf{x} = \lambda \mid |\mathsf{v}\rangle = |\xi_k\rangle] \Pr[|\mathsf{v}\rangle = |\xi_k\rangle] \\
&= \sum_{k=1}^{n} \left(\sum_{\lambda_i=\lambda} \langle\psi_i \mid \xi_k\rangle \langle\xi_k \mid \psi_i\rangle \right) p_k \\
&= \sum_{k=1}^{n} p_k \,\mathsf{Tr}\left(|\xi_k\rangle \langle\xi_k| \sum_{\lambda_i=\lambda} |\psi_i\rangle \langle\psi_i| \right) = \mathsf{Tr}(\rho \mathbf{E}_\lambda(\mathbf{O})).
\end{aligned}$$

Expected Value of an Observable

In the scenario that we just described let's find the expected value of the observed value x

$$\mathbb{E}\left[\mathbf{O}\right]_{\rho} = \mathbb{E}\left[\mathsf{x}\right] = \sum_{\lambda} \lambda \Pr\left[\mathsf{x} = \lambda\right] = \sum_{\lambda} \lambda \operatorname{Tr}(\rho(\mathbf{E}_{\lambda}(\mathbf{O}))) = \operatorname{Tr}(\rho\mathbf{O}),$$

and we notice that this quantity is linear in **O**.

In particular, if we were measuring a random observable,

$$\mathbf{O}_{\mathsf{o}} \sim \begin{pmatrix} \mathbf{O}_1 & \mathbf{O}_2 & \dots & \mathbf{O}_m \\ f_1 & f_2 & \dots & f_m \end{pmatrix},$$

we find that the order in which we take expectation commutes

$$\mathbb{E}\left[\mathbf{O}_{\mathsf{o}}\right] = \sum_{o=1}^{m} \mathbb{E}\left[\mathbf{O}_o\right] f_o = \mathbb{E}\left[\sum_{o} \mathbf{O}_o f_o\right],$$

i.e. **average expected value = expected value of average observable**

Observables as Quantum Channels

Depending on the true value of the initial state, $|\mathsf{v}\rangle$ may collapse into different states, yielding random state $|\mathsf{v}'\rangle$. With probability

$$q_{k,\lambda} = \sum_{k=1}^{n} \Pr\left[\mathsf{x} = \lambda \mid |\mathsf{v}\rangle = |\xi_k\rangle\right] \Pr\left[|\mathsf{v}\rangle = |\xi_k\rangle\right] = \mathsf{Tr}(|\xi_k\rangle \langle \xi_k| \, \mathbf{E}_\lambda(\mathbf{O})) p_k$$

we know that $|\mathsf{v}'\rangle \propto \mathbf{E}_\lambda(\mathbf{O}) \, |\xi_k\rangle$, with the proportionality constant chosen such that always $\left\| |\mathsf{v}'\rangle \right\|_2 = 1$ i.e.

$$\begin{aligned} C_{k,\lambda} =& 1/ \left\| \mathbf{E}_\lambda(\mathbf{O}) \, |\xi_k\rangle \right\|_2 = 1/\sqrt{\left\| \mathbf{E}_\lambda(\mathbf{O}) \, |\xi_k\rangle \right\|_2^2} \\ =& 1/\sqrt{\langle \xi_k| \, \mathbf{E}_\lambda(\mathbf{O})^\dagger \mathbf{E}_\lambda(\mathbf{O}) \, |\xi_k\rangle} = 1/\sqrt{\mathsf{Tr}\left(|\xi_k\rangle \langle \xi_k| \, \mathbf{E}_\lambda(\mathbf{O})\right)} = \sqrt{p_k / q_{k,\lambda}} \end{aligned}$$

so

$$\rho' = \sum_{k,\lambda} q_{k,\lambda} C_{k,\lambda}^2 \mathbf{E}_\lambda(\mathbf{O}) \, |\xi_k\rangle \langle \xi_k| \, \mathbf{E}_\lambda(\mathbf{O})^\dagger = \sum_{\lambda} \mathbf{E}_\lambda(\mathbf{O}) \rho \mathbf{E}_\lambda(\mathbf{O}).$$

Quantum Instruments

The problem with the previous view is that we are inherently discarding the newly-observed λ. To fully model an observable $\mathbf{O}$ as a quantum channel $\mathcal{E}_{\mathbf{O}}$, it should take as input a state $\rho \in \mathcal{B}(\mathbb{C}^d)$ and output both the post-measurement state $\rho_\lambda \in \mathcal{B}(\mathbb{C}^d)$ and also λ, which we model as $|\lambda\rangle\langle\lambda| \in \mathcal{B}(\mathcal{H})$ where $\dim \mathcal{H}$ is equal to the number of distinct eigenvalues of $\mathbf{O}$.

$$\begin{aligned}
\mathcal{E}_{\mathbf{O}}(\rho) &= \sum_{k,\lambda} q_{k,\lambda} C_{k,\lambda}^2 \mathbf{E}_\lambda(\mathbf{O}) |\xi_k\rangle\langle\xi_k| \mathbf{E}_\lambda(\mathbf{O}) \otimes |\lambda\rangle\langle\lambda| \\
&= \sum_{\lambda} \mathbf{E}_\lambda(\mathbf{O}) \rho \mathbf{E}_\lambda(\mathbf{O}) \otimes |\lambda\rangle\langle\lambda| \\
&= \sum_{\lambda} \mathbf{E}_\lambda(\mathbf{O}) \otimes |\lambda\rangle \rho \mathbf{E}_\lambda(\mathbf{O}) \otimes \langle\lambda| .
\end{aligned}$$

i.e. to model $\mathbf{O}$ as a quantum instrument we use a channel with Kraus operators $\{\mathbf{E}_\lambda(\mathbf{O}) \otimes |\lambda\rangle\}$. Notice that if we observe $\mathbf{O}$ on subsystem B of $|\psi\rangle_{AB} \in \mathcal{A} \otimes \mathbb{C}^d$, the full channel is $\mathbf{I}_A \otimes \mathcal{E}_{\mathbf{O}}$ with Kraus operators $\{\mathbf{I}_A \otimes \mathbf{E}_\lambda(\mathbf{O}) \otimes |\lambda\rangle\}$, as if we measured $\mathbf{I}_A \otimes \mathbf{O}$.

The CHSH Scenario

Let's say that Alice and Bob share a quantum state $\rho \in \mathcal{H}_A \otimes \mathcal{H}_B$ and they both would like to exploit it to answer a binary question.

For each such question that Alice may ask we assign an observable $\mathbf{A}_a \in \mathcal{B}(\mathcal{H}_A)$, and similarly Bob has his $\{\mathbf{B}_b\}$, and they can both choose what they want to ask of this quantum state.

Using the results of these questions they would like to figure out if the state they had shared exhibits quantum entanglement.

The setup they decide to use is as follows: both Alice and Bob can ask one of two questions (so a, b are bits), their observables only have ± 1 eigenvalues and the state is a qubit. They approximate

$$\mathcal{S} = \sum_{a,b} (-1)^{ab} \mathbb{E}\left[\mathbf{A}_a \otimes \mathbf{B}_b\right],$$

and notice that in general $|\mathcal{S}| \leq 2\sqrt{2}$, but if their state was not initally entagled then $|\mathcal{S}| \leq 2$.

Khalfin–Tsirelson–Landau Identity

In the CHSH scenario the weighted observable $\mathbf{S}$ satisfies

$$
\begin{aligned}
\mathbf{S}^2 =& \left(\sum_{a,b}(-1)^{ab}\mathbf{A}_a \otimes \mathbf{B}_b\right)^2 \\
=& \left(\mathbf{A}_0 \otimes \mathbf{B}_0 + \mathbf{A}_0 \otimes \mathbf{B}_1 + \mathbf{A}_1 \otimes \mathbf{B}_0 - \mathbf{A}_1 \otimes \mathbf{B}_1\right)^2 \\
=& \left(\mathbf{A}_0 \otimes (\mathbf{B}_0 + \mathbf{B}_1) + \mathbf{A}_1 \otimes (\mathbf{B}_0 - \mathbf{B}_1)\right)^2 \\
=& \mathbf{A}_0^2 \otimes (\mathbf{B}_0 + \mathbf{B}_1)^2 + \mathbf{A}_1^2 \otimes (\mathbf{B}_0 - \mathbf{B}_1)^2 \\
& + \mathbf{A}_0\mathbf{A}_1 \otimes (\mathbf{B}_0 + \mathbf{B}_1)(\mathbf{B}_0 - \mathbf{B}_1) + \mathbf{A}_1\mathbf{A}_0 \otimes (\mathbf{B}_0 - \mathbf{B}_1)(\mathbf{B}_0 + \mathbf{B}_1) \\
=& \mathbf{I} \otimes (\mathbf{B}_0^2 + \mathbf{B}_1^2 + \mathbf{B}_0\mathbf{B}_1 + \mathbf{B}_1\mathbf{B}_0) + \mathbf{I} \otimes (\mathbf{B}_0^2 + \mathbf{B}_1^2 - \mathbf{B}_0\mathbf{B}_1 - \mathbf{B}_0\mathbf{B}_1) \\
& + \mathbf{A}_0\mathbf{A}_1 \otimes (\mathbf{B}_0^2 - \mathbf{B}_1^2 + \mathbf{B}_1\mathbf{B}_0 - \mathbf{B}_0\mathbf{B}_1) + \mathbf{A}_1\mathbf{A}_0 \otimes (\mathbf{B}_0^2 - \mathbf{B}_1^2 + \mathbf{B}_0\mathbf{B}_1 - \mathbf{B}_1\mathbf{B}_0) \\
=& 4\mathbf{I} - (\mathbf{A}_0\mathbf{A}_1 - \mathbf{A}_1\mathbf{A}_0) \otimes (\mathbf{B}_0\mathbf{B}_1 - \mathbf{B}_1\mathbf{B}_0),
\end{aligned}
$$

noting that

$$
\mathbf{S}^2 \leq 4\mathbf{I} + |\mathbf{A}_0\mathbf{A}_1 - \mathbf{A}_1\mathbf{A}_0| \otimes |\mathbf{B}_0\mathbf{B}_1 - \mathbf{B}_1\mathbf{B}_0| \leq 8\mathbf{I},
$$

so expected value $\mathcal{S}$, which is less than the largest eigenvalue of $\mathbf{S}$, is at most $2\sqrt{2}$ (**Tsirelson Inequality**).

Sharpness of the Tsirelson Inequality

To have $\mathcal{S} = 2\sqrt{2}$ we need the pairs of observables to anticommute, i.e. $\mathbf{A}_0\mathbf{A}_1 = -\mathbf{A}_1\mathbf{A}_0$ and $\mathbf{B}_0\mathbf{B}_1 = -\mathbf{B}_1\mathbf{B}_0$.
One famous pair is $\{\mathbf{Z}, \mathbf{X}\}$, for which $\mathbf{ZX} - \mathbf{XZ} = 2\imath\mathbf{Y}$.
To fabricate another pair, take $\{\mathbf{UZU}^\dagger, \mathbf{UXU}^\dagger\}$ and notice

$$\mathbf{UZU}^\dagger\mathbf{UXU}^\dagger - \mathbf{UXU}^\dagger\mathbf{UZU}^\dagger = \mathbf{U}(\mathbf{ZX} - \mathbf{XZ})\mathbf{U}^\dagger = 2\imath\mathbf{UYU}^\dagger$$

so we e.g. take $\mathbf{A}_0 = \mathbf{Z}, \mathbf{A}_1 = \mathbf{X}, \mathbf{B}_0 = \mathbf{UZU}^\dagger, \mathbf{B}_1 = \mathbf{UXU}^\dagger$. Then

$$\begin{aligned}
\mathbf{S} =& (\mathbf{I} \otimes \mathbf{U}) (\mathbf{Z} \otimes \mathbf{Z} + \mathbf{Z} \otimes \mathbf{X} + \mathbf{X} \otimes \mathbf{Z} - \mathbf{X} \otimes \mathbf{X}) \left(\mathbf{I} \otimes \mathbf{U}^\dagger\right) \\
=& (\mathbf{I} \otimes \mathbf{U}) \begin{bmatrix} 1 & 1 & 1 & -1 \\ 1 & -1 & -1 & -1 \\ 1 & -1 & -1 & 1 \\ -1 & -1 & 1 & 1 \end{bmatrix} \left(\mathbf{I} \otimes \mathbf{U}^\dagger\right) \\
=& 2\sqrt{2} (\mathbf{I} \otimes \mathbf{U}) (|\phi_+\rangle \langle\phi_+| - |\phi_-\rangle \langle\phi_-|) \left(\mathbf{I} \otimes \mathbf{U}^\dagger\right),
\end{aligned}$$

where $|\phi_\pm\rangle \propto \begin{bmatrix} -1 & 1 \mp \sqrt{2} & 1 \mp \sqrt{2} & 1 \end{bmatrix}^\mathsf{T}$.
Example: take $\mathbf{U} = \mathbf{I}$ and initial shared state $|\phi_+\rangle \Rightarrow \mathcal{S} = 2\sqrt{2}$.

CHSH Inequality: Pedantry

For the final act, let's show that without entanglement $|\mathcal{S}| \leq 2$. For full generality we can consider a general probability space $(\Omega, \mathcal{F}, \mu)$ and the random variables $\mathsf{A}_a : \mathcal{F} \to \{\pm 1\}$ and $\mathsf{B}_b : \mathcal{F} \to \{\pm 1\}$ describing the ± 1-valued observation results of Alice and Bob, and use Lebesgue integrals to express $\mathbb{E}$. However, this is both tedious and pointless.

It is sufficient to look at $\mathcal{F}$ the Borel algebra of $\Omega = \mathbb{R}$, thus fixing the probability density function of the real hidden variable ω as

$$p(\omega) = \left. \frac{d}{dx} \mu((-\infty, x)) \right|_{x=\omega}$$

and define the functions $A_a : \mathbb{R} \to \{\pm 1\}$ and $B_b : \mathbb{R} \to \{\pm 1\}$ that indicate what each value of the hidden variable $\omega \in \Omega$ produces for Alice and Bob, i.e. the unique A_a and B_b such that

$$\mathsf{A}_a((-\infty, \omega)) = \int_{-\infty}^{\omega} A_a(\omega) p(\omega) d\omega.$$

CHSH Inequality: Setup

Considering the outcomes of $\mathbf{A}_a$ and $\mathbf{B}_b$ to be the result of stochastic actions applied to some classical shared real hidden variable ω we write

$$\mathbb{E}\left[\mathbf{A}_a \otimes \mathbf{B}_b\right] = \int_{\mathbb{R}} A_a(\omega) B_b(\omega) p(\omega) d\omega,$$

and we set out to prove that

$$2 \geq |\mathcal{S}| = \left|\mathbb{E}\left[\mathbf{A}_0 \otimes \mathbf{B}_0\right] + \mathbb{E}\left[\mathbf{A}_0 \otimes \mathbf{B}_1\right] + \mathbb{E}\left[\mathbf{A}_1 \otimes \mathbf{B}_0\right] - \mathbb{E}\left[\mathbf{A}_1 \otimes \mathbf{B}_1\right]\right|,$$

and we are going to follow Bell's original 1971 derivation, but backwards, so we can see how the sausage is made.

CHSH Inequality: Lucky Shot

We wanted to prove that

$$2 \geq |\mathcal{S}| = |\underbrace{\mathbb{E}\left[\mathbf{A}_0 \otimes \mathbf{B}_0\right] + \mathbb{E}\left[\mathbf{A}_0 \otimes \mathbf{B}_1\right]}_{x} + \underbrace{\mathbb{E}\left[\mathbf{A}_1 \otimes \mathbf{B}_0\right] - \mathbb{E}\left[\mathbf{A}_1 \otimes \mathbf{B}_1\right]}_{y}|.$$

To start, we want to split the sum on the right into two.
One thing we may think to use is that $|x + y| \leq |x| + |y|$,
but that doesn't give us a ceiling on $|\mathcal{S}|$ but a floor. In other words,

$$|\mathcal{S}| \leq |x| + |y|,$$

and if this was less than 2 we would be good, but this is, a priori, even harder to prove. But... what would we need to do it?

$$|x| + |y| \leq 2 \Leftrightarrow |y| \leq 2 - |x| \Leftrightarrow |y| \leq 2 - \max(x, -x) \Leftrightarrow |y| \leq 2 \pm x,$$

so we now need to prove two things, but we persevere.

CHSH Inequality: Two-For

We wanted to prove that

$$|\underbrace{\mathbb{E}\left[\mathbf{A}_1 \otimes \mathbf{B}_0\right] - \mathbb{E}\left[\mathbf{A}_1 \otimes \mathbf{B}_1\right]}_{y}| \leq 2 \pm \underbrace{\mathbb{E}\left[\mathbf{A}_0 \otimes \mathbf{B}_0\right] + \mathbb{E}\left[\mathbf{A}_0 \otimes \mathbf{B}_1\right]}_{x}.$$

and to get $\pm$ on the right-hand side we need to pull some tricks:

$$\begin{aligned}
y &= \int_R A_1(\omega)B_0(\omega)p(\omega)d\omega - \int_R A_1(\omega)B_1(\omega)p(\omega)d\omega \\
&= \int_R [A_1(\omega)B_0(\omega) \pm A_0(\omega)B_0(\omega)A_1(\omega)B_1(\omega) \\
&\quad - A_1(\omega)B_1(\omega) \mp A_0(\omega)B_0(\omega)A_1(\omega)B_1(\omega)\,]\,p(\omega)d\omega \\
&= \int_R \underbrace{A_1(\omega)B_0(\omega)}_{\in[-1,1]}\underbrace{(1 \pm A_0(\omega)B_1(\omega))}_{\geq 0}p(\omega)d\omega \\
&\quad - \int_R \underbrace{A_1(\omega)B_1(\omega)}_{\in[-1,1]}\underbrace{(1 \pm A_0(\omega)B_0(\omega))}_{\geq 0}p(\omega)d\omega
\end{aligned}$$

$$\Rightarrow |y| \leq \int_{\mathbb{R}}(1 \pm A_0(\omega)B_1(\omega))p(\omega)d\omega + \int_{\mathbb{R}}(1 \pm A_0(\omega)B_0(\omega))p(\omega)d\omega = 2 \pm x,$$

and we proved more than CHSH: $|\mathcal{S}| = |x + y| \leq |x| + |y| \leq 2$.